ALGEBRA 1 · UNIT 3: LINEAR INEQUALITIES

Aquarium Keeper

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students set up a 75-gallon classroom aquarium. They use compound inequalities to keep temperature and pH in safe ranges, find the overlap where different fish can live together, and use stocking and budget inequalities to design their own community tank.

This project emphasizes compound inequalities as ranges and the overlap (intersection) of ranges. Students see 'and' as the region where all conditions are true, and 'or' as the danger zone. The heater and evaporation problems connect inequalities to rates.

Standards Addressed

- A-CED.1 Create inequalities in one variable and use them to solve problems
- A-CED.3 Represent constraints with inequalities
- A-REI.3 Solve linear inequalities in one variable
- Honors: model ranges with absolute value inequalities

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a photo of a fish tank: What could go wrong? Part 1 safe ranges and overlaps.
Day 2	Part 2 stocking, heater, and evaporation inequalities.
Day 3	Part 3 design a community tank using the fish cards; write the care plan.

Materials

- Student handout and fish cards
- Calculators
- Colored pencils
- Optional: photos of a real classroom aquarium

How the Three Versions Differ

Level	What changes
CP2	Single species ranges, graphing compound inequalities, two-step stocking and heater inequalities with starters.
CP1	Finds the overlap of two species' ranges, converts to Celsius, compares stocking vs. budget to find the tighter constraint.
Honors	Uses absolute value to describe ranges, chooses a thermostat setting that keeps a tolerance band safe, and analyzes cooling.

Watch For (Common Misconceptions)

- Graphing the union instead of the overlap when two fish must both be safe.
- Rounding the number of fish up when the inequality says 'at most'.
- Writing $80 \le T \le 74$ (reversed order) for a compound inequality.

Extension Ideas

- Research real fish needs and build a class tank plan.
- Add pH adjustments (each dose changes pH by 0.2).
- Graph the heater model and shade the safe band (previews Unit 6).

Answer Key – CP2

Part 1: Safe Ranges

#	Task	Answer
1	Write a compound inequality for the safe temperature T. Then graph it.	$74 \le T \le 80$; closed dots at 74 and 80, shaded between.
2	Which of these thermometer readings are safe: 73°F, 76.5°F, 80°F, 81°F? Explain.	76.5°F and 80°F are safe. 73°F is too cold and 81°F is too warm.
3	Write a compound inequality for the safe pH level p.	$6.5 \le p \le 7.5$
4	Write an inequality for the temperatures that are NOT safe.	$T < 74$ or $T > 80$

Part 2: Fish, Heaters & Water

#	Task	Answer
5	The tank already has 6 angelfish that are 6 inches long (36 inches of fish). How many 1.5-inch neon tetras can you add without breaking the…	$36 + 1.5n \le 75 \rightarrow 1.5n \le 39 \rightarrow n \le 26$ tetras
6	The water starts at 68°F. The heater warms it 1.5°F every hour. After how many hours h is the water at least 74°F?	$68 + 1.5h \ge 74 \rightarrow 1.5h \ge 6 \rightarrow h \ge 4$ hours
7	Write and solve a compound inequality to find the hours when the water is in the safe range: $74 \le 68 + 1.5h \le 80$.	$6 \le 1.5h \le 12 \rightarrow 4 \le h \le 8$
8	When should the class turn the heater down? Explain using your answer.	By 8 hours, because after 8 hours the water would be warmer than 80°F.

Part 3: Design Your Tank

#	Task	Answer
9	Choose two kinds of fish. Write each one's temperature range as a compound inequality.	Answers vary. Example: Neon $72 \le T \le 80$; Guppy $74 \le T \le 82$.
10	Find the temperatures that are safe for BOTH fish. Write one compound inequality.	Example: Neon and Guppy → $74 \le T \le 80$.
11	Choose how many of each fish to buy. Show that the total length is at most 75 inches.	Answers vary; check that total length ≤ 75 inches.

Answer Key – CP1

Part 1: Safe Ranges & Overlaps

#	Task	Answer
1	Write compound inequalities for the safe ranges of neon tetras and angelfish.	Neon: $72 \le T \le 80$. Angelfish: $76 \le T \le 84$.
2	Write one compound inequality for temperatures that are safe for both species. Graph it.	$76 \le T \le 80$; closed dots at 76 and 80.
3	The tank's digital thermometer reads Celsius, and $F = 1.8C + 32$. Write and solve a compound inequality for the safe Celsius range for both…	$76 \le 1.8C + 32 \le 80 \rightarrow 44 \le 1.8C \le 48 \rightarrow 24.4 \le C \le 26.7$
4	Write an inequality for the temperatures that would be unsafe for at least one of the two species.	$T < 76$ or $T > 80$

Part 2: Stocking, Heaters & Budget

#	Task	Answer
5	Write and solve a stocking inequality for the number of neon tetras n.	Existing: $2(6) + 6(2.5) = 27$ in. $27 + 1.5n \le 75 \rightarrow n \le 32$ tetras
6	The water starts at 68°F and the heater adds 1.5°F per hour. When is the water safe for both species? Give your answer in hours and minutes.	$76 \le 68 + 1.5h \le 80 \rightarrow 8 \le 1.5h \le 12 \rightarrow 5.33 \le h \le 8 \rightarrow$ from 5 h 20 min to 8 h
7	The tank loses 0.5 gallon per day to evaporation and must stay at or above 70 gallons. Within how many days must it be refilled?	$75 - 0.5d \ge 70 \rightarrow -0.5d \ge -5 \rightarrow d \le 10$ days
8	Your budget for fish is $120. Corydoras cost $4 each and tetras cost $2.50 each (the angelfish are already paid for). How many tetras can you…	$24 + 2.50n \le 120 \rightarrow n \le 38.4 \rightarrow 38$ tetras. Stocking (32) is more limiting than budget (38).

Part 3: Design a Community Tank

#	Task	Answer
9	List your species and find the overlap of all their temperature ranges as one compound inequality. If there is no overlap, change a species.	Answers vary; check the overlap (the greatest lower bound to the least upper bound).
10	Write and solve stocking and budget inequalities for your tank. Show that your final numbers satisfy both.	Answers vary.

11	Write a maintenance schedule using at least two inequality results (heater time, refill days, safe readings).	Answers vary.

Answer Key – Honors

Part 1: Ranges as Distances

#	Task	Answer
1	Write the safe ranges for neon tetras (72–80°F) and angelfish (76–84°F) as absolute value inequalities.	Neon: $\|T - 76\| \le 4$. Angelfish: $\|T - 80\| \le 4$.
2	Find the overlap of the two ranges. Write it as a compound inequality and as an absolute value inequality.	$76 \le T \le 80$; $\|T - 78\| \le 2$
3	Convert the overlap to Celsius using $F = 1.8C + 32$ (round to the nearest hundredth). Then write it as an absolute value inequality in C.	$24.44 \le C \le 26.67$; center ≈ 25.56, radius ≈ 1.11 → $\|C - 25.56\| \le 1.11$
4	The safe pH range is 6.5 to 7.5. Write it as an absolute value inequality. A test reads 7.6. By how much is it outside the range?	$\|p - 7\| \le 0.5$. 7.6 is 0.1 above the maximum of 7.5.

Part 2: Thermostats, Cooling & Stocking

#	Task	Answer
5	The thermostat keeps the water within 0.8°F of its setting s, so $\|T - s\| \le 0.8$. Find every setting s that keeps the water safe. Explain your…	Need $s - 0.8 \ge 76$ and $s + 0.8 \le 80$ → $76.8 \le s \le 79.2$
6	If the power goes out at night, the water cools 0.4°F per hour starting from 79°F. After how many hours is the water unsafe?	$79 - 0.4h < 76$ → $-0.4h < -3$ → $h > 7.5$ hours
7	The tank loses 0.6 gallon per day and must stay at least 70 gallons. Find the longest whole number of days between refills.	$75 - 0.6d \ge 70$ → $d \le 8.33$ → 8 days
8	The tank has 2 angelfish (6 in each) and 6 corydoras (2.5 in each). Tetras must be added in a school of at least 6. The fish budget is $120 (only…	Stocking: $27 + 1.5n \le 75$ → $n \le 32$. Budget: $24 + 2.50n \le 120$ → $n \le 38.4$. Schooling: $n \ge 6$. So $6 \le n \le 32$.

Part 3: Engineer a Community Tank

#	Task	Answer
9	Write each species' range as an absolute value inequality. Find the overlap of all of them and the thermostat settings that keep a 0.8°F tolerance…	Answers vary; settings: (lower bound + 0.8) ≤ s ≤ (upper bound − 0.8).
10	Write and solve stocking and budget inequalities. Choose final counts and prove they satisfy every constraint.	Answers vary.
11	Write a care plan that explains what to do if the power goes out, using an inequality to give a time limit.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Chosen fish share a common temperature range and the stocking and budget inequalities are satisfied. Goes beyond the minimum with a creative, realistic choice.	Chosen fish share a common temperature range and the stocking and budget inequalities are satisfied.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Care plan includes the fish list, every safe range as an inequality, graphs of the overlap, and a maintenance schedule. Polished, organized, and visually engaging.	Care plan includes the fish list, every safe range as an inequality, graphs of the overlap, and a maintenance schedule. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.